1 Sequences

Definition 1.1 (Sequence)**.** A *sequence* is an ordered list of real numbers $a_1, a_2, a_3, \ldots$. Each number a_n is called a *term* of the sequence. The subscript n is called the *index*.

More formally, a sequence can be thought of as a *function* $a : \mathbb{N} \to \mathbb{R}$, where we write $a(n) = a_n$. We often denote a sequence by $\{a_n\}_{n=1}^{\infty}$ or just $\{a_n\}$.

Example 1.2. The sequence $1, 4, 7, 10, \ldots$ can be defined in two ways:

(i) It can be defined by a recursive formula:

$$\begin{aligned} a_1 &= 1, \\ a_{n+1} &= a_n + 3 \end{aligned}$$

(ii) It can also be defined by an explicit formula for the nth term:

$$\begin{aligned} a_n &= 1 + 3(n-1) \\ &= 3n - 2, \quad n = 1, 2, 3, \ldots. \end{aligned}$$

Definition 1.3 (Convergence)**.** A sequence $\{a_n\}$ is said to *converge* to a real number L if given any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|a_n - L| < \varepsilon$. In this case, we write $\lim_{n\to\infty} a_n = L$. If a sequence does not converge, it is said to *diverge*. If given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $a_n > \varepsilon$, we say that $\{a_n\}$ *diverges to infinity* and write $\lim_{n\to\infty} a_n = \infty$.

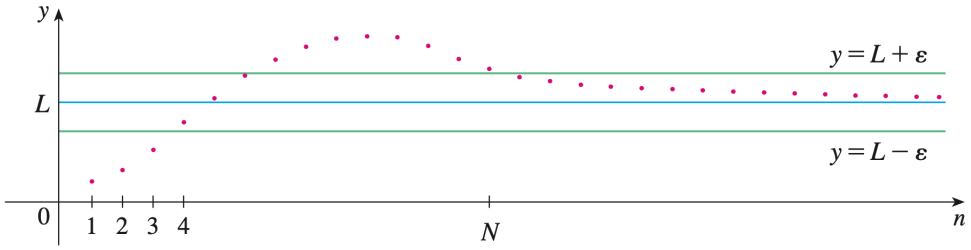
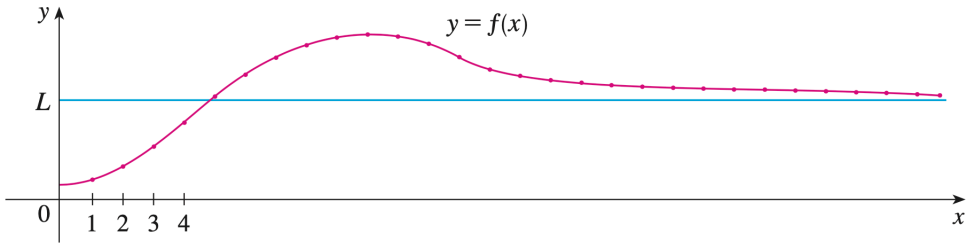


Figure 1: A convergent sequence.

Notice that the only difference between the definitions of $\lim_{n\to\infty} a_n = L$ and $\lim_{x\to\infty} f(x) = L$ is that n is required to be an integer. Thus, we have the following theorem:

Name:_____

Theorem 1.4. *If* $\lim_{x\to\infty} f(x) = L$ *and* $f(n) = a_n$ *for all* $n \in \mathbb{N}$, *then* $\lim_{n\to\infty} a_n = L$.



1. Prove Theorem 1.4 in the case where $L \in \mathbb{R}$. (Note that Theorem 1.4 also holds for infinite limits.)

2. Use Proposition (1.4) to prove that the sequence $a_n = \frac{1}{n}$ converges and find the limit of the sequence.

3. Find $\lim\limits_{n\to\infty} \dfrac{n}{n+1}$, if it exists.

4. Find $\lim_{n\to\infty}(-1)^n$, if it exists.

5. Find $\lim_{n\to\infty} \dfrac{n}{\sqrt{10+n}}$.

6. Find $\lim_{n\to\infty} \dfrac{\ln(n)}{n}$, if it exists.

2 Limit Laws for Sequences

We have the following limit laws for sequences.

Theorem 2.1 (Limit Laws for Sequences)**.** *If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and c is a constant, then*

(1) $\lim_{n\to\infty}(a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$

(2) $\lim_{n\to\infty}(ca_n) = c\lim_{n\to\infty} a_n$

(3) $\lim_{n\to\infty} c = c$

(4) $\lim_{n\to\infty}(a_n b_n) = \lim_{n\to\infty} a_n \lim_{n\to\infty} b_n$

(5) $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$, *provided that* $\lim_{n\to\infty} b_n \neq 0$.

(6) $\lim_{n\to\infty} a_n^p = [\lim_{n\to\infty} a_n]^p$ *if* $p > 0$ *and* $a_n > 0$

The proofs of these are similar to the those for functions. Here is an example.

Example 2.2 (Proof of Limit Law 1)**.** Let $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$. Given $\varepsilon > 0$, there exist positive integers N_1 and N_2 such that $n \geq N_1$ implies $|a_n - L| < \frac{\varepsilon}{2}$ and $n \geq N_2$ implies $|b_n - L| < \frac{\varepsilon}{2}$. It follows that $n \geq \max\{N_1, N_2\}$ implies

$$|a_n + b_n - (L+M)| = |a_n - L + b_m - M| \leq |a_n - L| + |b_m - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which proves that $\lim_{n\to\infty}(a_n + b_n) = L + M = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$.

5. Prove Limit Law (2).

The Squeeze Theorem also holds for sequences.

Theorem 2.3 (Squeeze Theorem for Sequences)**.** *If* $a_n \leq b_n \leq c_n$ *for* $n \geq N$ *and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$*, then* $\lim_{n\to\infty} b_n = L$*.*

The proof is again essentially the same as before.

6. Find $\lim_{n\to\infty} n^2 \sin\left(\frac{1}{n}\right)$, if it exists.

7. Use the Squeeze Theorem to prove that $\lim_{n\to\infty} \frac{n!}{n^n} = 0$.

Theorem 2.4. *If* $\lim_{n\to\infty} |a_n| = 0$*, then* $\lim_{n\to\infty} a_n = 0$*.*

8. Prove Theorem 2.4. (Either use the definition or the Squeeze Theorem.)

9. Use Theorem 2.4 to find $\lim_{n\to\infty} \frac{(-1)^n}{n}$.

Theorem 2.5. *If* $\lim_{n\to\infty} a_n = L$ *and a function* f *is continuous at L, then*

$$\lim_{n\to\infty} f(a_n) = f(L).$$

10. Prove Theorem 2.5.

11. Find $\lim_{n\to\infty} \sin\left(\frac{\pi}{n}\right)$.

12. Let $a > 0$. Determine whether the sequence $\left\{\left(1+\frac{a}{n}\right)^n\right\}$ converges. If so, compute the limit.
Hint: $\lim_{n\to\infty}\left(1+\frac{a}{n}\right)^n = \lim_{n\to\infty}\exp\left(\ln\left(1+\frac{a}{n}\right)^n\right)$.

3 Geometric Sequences

13. For which values of r is the sequence $\{r^n\}$ convergent?

14. Determine whether each of the following sequences converges. In the case of convergence, determine the limit.

    (a) $a_n = 0.75^n$

    (b) $a_n = (-0.75)^n$

    (c) $a_n = 1.15^n$

    (d) $a_n = (-1.15)^n$